

[Vitamin K: Fact Sheet for Consumers \(original version\)](#).

Vitamin K: Fact Sheet for Consumers

Table of Contents

- [What is vitamin K and what does it do?](#)
- [How much vitamin K do I need?](#)
- [What foods provide vitamin K?](#)
- [What kinds of vitamin K dietary supplements are available?](#)
- [Am I getting enough vitamin K?](#)
- [What happens if I don't get enough vitamin K?](#)
- [What are some effects of vitamin K on health?](#)
- [Can vitamin K be harmful?](#)
- [Does vitamin K interact with medications or other dietary supplements?](#)
- [Vitamin K and healthful eating](#)
- [Where can I find out more about Vitamin K?](#)
- [Disclaimer](#)
- [Glossary](#)

What is vitamin K and what does it do?

Vitamin K is a nutrient that the body needs to stay healthy. It's important for blood clotting and healthy bones and also has other functions in the body. If you are taking a blood thinner such as warfarin (Coumadin), it's very important to get about the same amount of vitamin K each day.

How much vitamin K do I need?

The amount of vitamin K you need depends on your age and sex. Average daily recommended amounts are listed below in micrograms (mcg).

Life Stage	Recommended Amount
Birth to 6 months	2.0 mcg
7–12 months	2.5 mcg
1–3 years	30 mcg
4–8 years	55 mcg
9–13 years	60 mcg
14–18 years	75 mcg
Adult men 19 years and older	120 mcg
Adult women 19 years and older	90 mcg
Pregnant or breastfeeding teens	75 mcg
Pregnant or breastfeeding women	90 mcg

What foods provide vitamin K?

Vitamin K is found naturally in many foods. You can get recommended amounts of vitamin K by eating a variety of foods, including the following:

- Green leafy vegetables, such as spinach, kale, broccoli, and lettuce
- Vegetable oils

- Some fruits, such as blueberries and figs
- Meat, cheese, eggs, and soybeans

You can find links to more food sources of vitamin K in the last section of this fact sheet, **Where can I find out more about Vitamin K?**

What kinds of vitamin K dietary supplements are available?

Vitamin K is found in multivitamin/mineral supplements. Vitamin K is also available in supplements of vitamin K alone or of vitamin K with a few other nutrients such as calcium, magnesium, and/or vitamin D. Common forms of vitamin K in dietary supplements are phylloquinone and phytonadione (also called vitamin K1), menaquinone-4, and menaquinone-7 (also called vitamin K2).

Am I getting enough vitamin K?

Vitamin K deficiency is very rare. Most people in the United States get enough vitamin K from the foods they eat. Also, bacteria in the colon make some vitamin K that the body can absorb. However, certain groups of people may have trouble getting enough vitamin K:

- Newborns who don't receive an injection of vitamin K at birth
- People with conditions (such as cystic fibrosis, celiac disease, ulcerative colitis, and short bowel syndrome) that decrease the amount of vitamin K their body absorbs
- People who have had bariatric (weight loss) surgery

What happens if I don't get enough vitamin K?

Severe vitamin K deficiency can cause bruising and bleeding problems because the blood will take longer to clot. Vitamin K deficiency might reduce bone strength and increase the risk of getting osteoporosis because the body needs vitamin K for healthy bones.

What are some effects of vitamin K on health?

Scientists are studying vitamin K to understand how it affects our health. Here are some examples of what this research has shown.

Osteoporosis

Vitamin K is important for healthy bones. Some research shows that people who eat more vitamin K-rich foods have stronger bones and are less likely to break a hip than those who eat less of these foods. A few studies have found that taking vitamin K supplements improves bone strength and lessens the chances of breaking a bone, but other studies have not. More research is needed to better understand if vitamin K supplements can help improve bone health and reduce osteoporosis risk.

Coronary heart disease

Scientists are studying whether low blood levels of vitamin K increase the risk of coronary heart disease, perhaps by making blood vessels that feed the heart stiffer and narrower. More research is needed to understand whether vitamin K supplements might help prevent heart disease.

Can vitamin K be harmful?

Vitamin K has not been shown to cause any harm. However, it can interact with some medications, particularly warfarin (Coumadin); see the information below.

Does vitamin K interact with medications or other dietary supplements?

Yes, some medications may interact with vitamin K. Here are a few examples:

Warfarin (Coumadin)

Vitamin K can have a serious interaction with the blood thinner warfarin (Coumadin). If you take warfarin, make sure that the amount of vitamin K you consume from food and supplements is about the same every day. A sudden change in the amount of vitamin K you get can cause dangerous bleeding (if you consume less) or blood clots (if you consume more).

Antibiotics

Antibiotics can destroy the good bacteria in your gut. Some of these bacteria make vitamin K. Using antibiotics for more than a few weeks may reduce the amount of vitamin K made in your gut and, therefore, the amount available for your body to use.

Bile acid sequestrants

Some people take bile acid sequestrants (such as cholestyramine [Questran] and colestipol [Colestid]) to lower blood cholesterol levels. These medications can reduce the amount of vitamin K your body absorbs, especially if you take them for many years.

Orlistat

Orlistat (Alli and Xenical) is a weight-loss drug. It reduces the amount of fat your body absorbs and can decrease the absorption of vitamin K.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients such as vitamin K.

Vitamin K and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about Vitamin K?

- For general information on vitamin K
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Vitamin K](#)
 - [Vitamin K](#), MedLinePlus
- For more information on food sources of vitamin K
 - ODS Health Professional Fact Sheet on [Vitamin K](#)
 - USDA's [FoodData Central](#)
 - USDA Nutrient Lists for vitamin K as either
 - Phylloquinone, the form of vitamin K found in green leafy vegetables, listed by [food](#) or by [content](#), or

- Menaquinone, a form of vitamin K found in fermented foods and foods that come from animals, listed by [food](#) or by [content](#)
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

antibiotic

A drug used to treat infections caused by bacteria and other microorganisms.

bacteria

Single-celled organisms that are too small to be seen without a microscope. Bacteria are found everywhere and may be helpful or harmful.

bile acid sequestrant

A type of medication that is used to treat high cholesterol.

blood vessel

A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

calcium

A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

celiac disease

An autoimmune disorder in which eating gluten (a protein found in wheat, rye, barley, and possibly oats) causes the immune system to damage the small intestine, making it unable to absorb nutrients. It is a genetic disease that sometimes becomes active for the first time after surgery, pregnancy, childbirth, viral infection, or extreme stress. Also called sprue.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

colon

A tube-like organ about 5 feet long in adults that is connected to the small intestine at one end and the anus at the other. The colon absorbs water, some nutrients, and electrolytes (such as sodium and chloride) from partially digested food. The remaining material (solid waste called stool) moves through the colon to the rectum and leaves the body through the anus as a bowel movement. The colon is part of the digestive system (a series of organs from the mouth to the anus). Also called the large intestine.

consume

To eat or drink.

coronary heart disease

A disease in which the blood vessels (coronary arteries) that carry blood and oxygen to the heart are narrowed or blocked, which can cause chest pain, shortness of breath, and heart attack. It is usually caused by a build-up of fat and cholesterol deposits inside the arteries (atherosclerosis).

cystic fibrosis

A common inherited disease that causes the body to make thick, sticky mucus that builds up in the lungs and blocks the airways, leading to repeated serious lung infections. Mucus also blocks the pancreas, which stops digestive enzymes from reaching the intestines. Cystic fibrosis also causes very salty sweat, which can lead to dehydration, increased heart rate, tiredness, low blood pressure, and heat stroke.

deficiency

An amount that is not enough; a shortage.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

injection

Use of a syringe and needle to push fluids or drugs into the body. Also called a shot.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

magnesium

In nutrition, a mineral the body needs for normal muscles, nerves, and bones. It also helps keep a steady heart rhythm, a healthy immune system, normal blood sugar levels and blood pressure, and is involved in making energy and protein for the body. Magnesium is found in some foods, including green vegetables, beans and peas, nuts and seeds, and whole grains.

microgram

μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

osteoporosis

A condition in which bones become weak and brittle, increasing the chance they may break.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

short bowel syndrome

A condition in which people cannot absorb enough water, vitamins, minerals, protein, fat, calories, and other nutrients from the food they eat because their small intestine is too short. The intestine is too short because they have undergone surgery to remove a portion of it, it is damaged, or they were born without one.

soy

A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

ulcerative colitis

Chronic inflammation of the colon that causes ulcers to form in its lining. This condition is marked by abdominal pain, cramps, and loose discharges of pus, blood, and mucus from the bowel.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin D

A nutrient that is obtained from the diet and can be made in the skin after exposure to sunlight. Vitamin D acts as a hormone. It helps to form and maintain strong bones, maintain normal blood levels of calcium and phosphorus, and increase calcium absorption; it also helps to maintain a healthy immune system and control cell growth. Vitamin D is found in some foods, including some types of fatty fish, and milk and breakfast cereals that are fortified with vitamin D.

vitamin K

A nutrient needed by the body to function and stay healthy. It helps form blood clots and maintain strong bones. Vitamin K is found in some foods, including green leafy vegetables, broccoli, liver, and vegetable oils. It is also made by bacteria that live in the large intestine.

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